

Problem 2

Part a)

Step 1: Moments of X_1 .

Since $X_1 \sim U(-1, 1)$, it has density $f(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x)$ (Example 1.5.6a). We compute the first three moments of X_1 :

$$\begin{aligned}\mathbb{E}(X_1) &= \frac{1}{2} \int_{-1}^1 x dx \stackrel{(\text{symmetry})}{=} 0, \\ \mathbb{E}(X_1^2) &= \frac{1}{2} \int_{-1}^1 x^2 dx = \frac{1}{2} \cdot \frac{x^3}{3} \Big|_{-1}^1 = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}, \\ \mathbb{E}(X_1^3) &= \frac{1}{2} \int_{-1}^1 x^3 dx \stackrel{(\text{symmetry})}{=} 0.\end{aligned}$$

We also have that:

$$\text{Var}(X_1) = \mathbb{E}(X_1^2) - (\mathbb{E}(X_1))^2 = \frac{1}{3} - 0 = \frac{1}{3} > 0.$$

Step 2: Reformulation of $\sqrt{n}Y_n$. Multiplying numerator and denominator of $\sqrt{n}Y_n$ by $\frac{1}{n}$ gives:

$$\sqrt{n}Y_n = \frac{\frac{\sqrt{n} \sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2 + \sum_{i=1}^n X_i^3}}{\frac{1}{n} \sum_{i=1}^n X_i^2 + \frac{1}{n} \sum_{i=1}^n X_i^3} = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i}{\frac{1}{n} \sum_{i=1}^n X_i^2 + \frac{1}{n} \sum_{i=1}^n X_i^3}.$$

We define:

$$U_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i, \quad V_n := \frac{1}{n} \sum_{i=1}^n X_i^2 + \frac{1}{n} \sum_{i=1}^n X_i^3,$$

so that $\sqrt{n}Y_n = U_n/V_n$. Rewriting the denominator:

$$\sum_{i=1}^n X_i^2 + \sum_{i=1}^n X_i^3 = \sum_{i=1}^n X_i^2(1 + X_i),$$

Since $X_i \sim U(-1, 1)$ is a continuous random variable, $X_i > -1$ a.s., so $1 + X_i > 0$ a.s. Moreover, $\mathbb{P}(X_i = 0) = 0$ for each $i \in \{1, \dots, n\}$ (continuous distribution), so $X_i^2 > 0$ a.s. for each $i \in \{1, \dots, n\}$. each summand satisfies $X_i^2(1 + X_i) > 0$ a.s., so $V_n > 0$ a.s. and Y_n is "well-defined a.s." (i.e., the event $\{V_n = 0\}$ has probability zero).

Step 3: Central Limit Theorem applied to U_n . By assumption, the random variables $(X_i)_{i \in \mathbb{N}}$ are independent and identically distributed, and $X_i \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ since $\mathbb{E}(X_1^2) = \frac{1}{3} < \infty$. By the Central Limit Theorem (Theorem 4.4.1) with $\mu = \mathbb{E}(X_1) = 0$ and $\sigma^2 = \text{Var}(X_1) = \frac{1}{3}$:

$$S_n^* := \frac{\sum_{i=1}^n X_i - n \cdot 0}{\sqrt{\frac{1}{3}} \cdot \sqrt{n}} = \frac{\sqrt{3}}{\sqrt{n}} \sum_{i=1}^n X_i \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0, 1). \quad (1)$$

We note that $U_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i = \frac{1}{\sqrt{3}} S_n^*$. To apply Slutsky's Theorem, define the sequences

$$\tilde{X}_n := S_n^*, \quad \tilde{Y}_n \equiv \frac{1}{\sqrt{3}}, \quad \tilde{Z}_n \equiv 0.$$

We have $\tilde{X}_n \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$ by (1). For any $\varepsilon > 0$, the event $\left\{ \left| \tilde{Y}_n - \frac{1}{\sqrt{3}} \right| > \varepsilon \right\} = \emptyset$, thus $\mathbb{P}(\emptyset) = 0$ (Lemma 1.1.10 a), so $\tilde{Y}_n \xrightarrow{p} \frac{1}{\sqrt{3}} =: y$ (Definition 4.1.1). By the same argument, $\tilde{Z}_n \xrightarrow{p} 0 =: z$. By Slutsky's Theorem, we have:

$$U_n = \tilde{Z}_n + \tilde{X}_n \tilde{Y}_n \xrightarrow{d} z + y Z = 0 + \frac{1}{\sqrt{3}} Z =: W.$$

Since $W = Z/\sqrt{3}$ is an affine transformation of a standard normal variable with $a = 1/\sqrt{3}$ and $b = 0$ (Example 3.1.15), we have:

$$W \sim \mathcal{N}\left(0, \frac{1}{3}\right).$$

Step 4: Law of Large Numbers applied to the partial sums of V_n . Since $|X_i| < 1$ a.s. for every $i \in \mathbb{N}$, the random variables X_i^2 and X_i^3 are bounded and hence in $L^1(\Omega, \mathcal{F}, \mathbb{P})$. Because $(X_i)_{i \in \mathbb{N}}$ are i.i.d. and $g_1(x) := x^2$, $g_2(x) := x^3$ are Borel-measurable functions, Lemma 3.1.9 guarantees that X_i^2 and X_i^3 are again random variables for every $i \in \mathbb{N}$. The sequences $(X_i^2)_{i \in \mathbb{N}}$ and $(X_i^3)_{i \in \mathbb{N}}$ are identically distributed since all X_i share the same distribution. They are independent because, for every finite $J \subset \mathbb{N}$ and measurable sets $(B_j)_{j \in J}$,

$$\mathbb{P}\left(\bigcap_{j \in J} \{X_j^2 \in B_j\}\right) = \mathbb{P}\left(\bigcap_{j \in J} \{X_j \in g_1^{-1}(B_j)\}\right) \stackrel{(\text{i.i.d.})}{=} \prod_{j \in J} \mathbb{P}(X_j \in g_1^{-1}(B_j)) = \prod_{j \in J} \mathbb{P}(X_j^2 \in B_j),$$

Hence $(X_i^2)_{i \in \mathbb{N}}$ is i.i.d. (Definition 3.2.5). The same argument applies to $(X_i^3)_{i \in \mathbb{N}}$. Since we also had that $(X_i^2)_{i \in \mathbb{N}}, (X_i^3)_{i \in \mathbb{N}} \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, we can apply Kolmogorov's Strong Law of Large Numbers (Theorem 4.3.6) to both sequences:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n X_i^2 &\xrightarrow{\text{a.s.}} \mathbb{E}(X_1^2) = \frac{1}{3}, \\ \frac{1}{n} \sum_{i=1}^n X_i^3 &\xrightarrow{\text{a.s.}} \mathbb{E}(X_1^3) = 0. \end{aligned} \tag{2}$$

Step 5: Convergence of V_n .

Question: Is it necessary to derive this from first principles? *I cannot find any statement in the lecture notes that states that the sum of almost surely convergent sequences converges almost surely.*

According to Definition 3.3.10, $\mathbb{P}$ -a.s. convergence means that the events where pointwise convergence holds have probability 1. Define these events as:

$$\begin{aligned} A &:= \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i^2(\omega) = \frac{1}{3} \right\}, \\ B &:= \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i^3(\omega) = 0 \right\}, \end{aligned}$$

where $\mathbb{P}(A) = 1$ and $\mathbb{P}(B) = 1$. We must show that $\mathbb{P}(A \cap B) = 1$ (i.e., the event where these limits hold simultaneously almost surely). Applying subadditivity (Lemma 1.1.10e) to the complement events, we obtain:

$$\mathbb{P}((A \cap B)^c) = \mathbb{P}(A^c \cup B^c) \leq \mathbb{P}(A^c) + \mathbb{P}(B^c) = 0 + 0 = 0.$$

Therefore, $\mathbb{P}(A \cap B) = 1$. Furthermore, the limit of a sum of convergent real sequences is the sum of their limits, so for all $\omega \in A \cap B$:

$$\begin{aligned} \lim_{n \rightarrow \infty} V_n(\omega) &= \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n X_i^2(\omega) + \frac{1}{n} \sum_{i=1}^n X_i^3(\omega) \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i^2(\omega) + \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i^3(\omega) \\ &= \frac{1}{3} + 0 = \frac{1}{3}. \end{aligned}$$

Since $\lim_{n \rightarrow \infty} V_n(\omega) = 1/3$ holds for all $\omega \in A \cap B$, and $\mathbb{P}(A \cap B) = 1$, then $V_n \xrightarrow{\text{a.s.}} 1/3$ (Definition 3.3.10). Since $\mathbb{P}$ -a.s. convergence implies convergence in probability (Theorem 4.1.2a), we also have $V_n \xrightarrow{p} 1/3$.

Step 6: Conclusion via Slutsky's Theorem. We have $U_n \xrightarrow{d} W \sim \mathcal{N}(0, \frac{1}{3})$ and $V_n \xrightarrow{p} \frac{1}{3} \neq 0$. Since $V_n > 0$ a.s., the indicator $\mathbb{I}_{\{V_n \neq 0\}} = 1$ a.s., so $\sqrt{n}Y_n = U_n/V_n$ a.s. Applying Slutsky's Theorem with $X_n = U_n$, $Y_n = V_n$, $Z_n \equiv 0$, $y = \frac{1}{3} \neq 0$, $z = 0$:

$$\sqrt{n}Y_n = 0 + \frac{U_n}{V_n} \mathbb{I}_{\{V_n \neq 0\}} \xrightarrow{d} 0 + \frac{W}{1/3} = 3W.$$

Since $W \sim \mathcal{N}(0, \frac{1}{3})$ and $3W$ is an affine transformation of W with $A = 3$, $b = 0$ (Example 3.1.15), we have:

$$3W \sim \mathcal{N}\left(3 \cdot 0, 3^2 \cdot \frac{1}{3}\right) = \mathcal{N}(0, 3).$$

Therefore, $\sqrt{n}Y_n$ converges **in distribution** to $\mathcal{N}(0, 3)$:

$$\sqrt{n}Y_n \xrightarrow{d} \mathcal{N}(0, 3).$$

□

Part b)

Define the sequence of deterministic constant random variables:

$$Y_n \equiv \frac{1}{\sqrt{n}}, \quad n \in \mathbb{N}.$$

Since Y_n is a deterministic sequence converging to 0, it converges to 0 in probability (Definition 4.1.1): for every $\varepsilon > 0$,

$$\mathbb{P}(|Y_n - 0| > \varepsilon) = \mathbb{P}\left(\frac{1}{\sqrt{n}} > \varepsilon\right) = 0 \quad \text{for all } n \text{ large enough,}$$

so $Y_n \xrightarrow{p} 0$. By hypothesis, $Z_n \xrightarrow{d} Z$. Applying Slutsky's Theorem, we obtain:

$$Z_n \cdot Y_n \xrightarrow{d} Z \cdot 0 = 0.$$

Computing the left-hand side explicitly:

$$Z_n \cdot Y_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n (X_k - \mu) \cdot \frac{1}{\sqrt{n}} = \frac{1}{n} \sum_{k=1}^n (X_k - \mu).$$

Hence

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \xrightarrow{d} 0. \quad (3)$$

Let $W \equiv 0$ be a random variable that is $\mathbb{P}$ -a.s. constant. Since our sequence converges in distribution to W , Theorem 4.1.8b) implies that it also converges in probability to W . Therefore,

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \xrightarrow{p} 0,$$

which is equivalent to

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{p} \mu. \quad (4)$$

Additionally, since $(X_k)_{k \in \mathbb{N}} \subset L^1(\Omega, \mathcal{F}, \mathbb{P})$ is i.i.d., Kolmogorov's Strong Law of Large Numbers (Theorem 4.3.6) gives

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{\text{a.s.}} \mathbb{E}(X_1).$$

By Theorem 4.1.2 a), $\mathbb{P}$ -almost-sure convergence implies convergence in probability, so

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{p} \mathbb{E}(X_1). \quad (5)$$

The limit in probability to a constant is unique. Because (4) and (5) show that the same sequence converges in probability to both μ and $\mathbb{E}(X_1)$, we must have:

$$\mathbb{E}(X_1) = \mu. \quad (6)$$

Substituting $\mathbb{E}(X_1) = \mu$ back into the Kolmogorov's Strong Law (Theorem 4.3.6), we immediately obtain:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = \mu \quad \mathbb{P}\text{-a.s.}$$

□

Part c)

Step 1: Algebraic reformulation. Let $S_n := \sum_{k=1}^n X_k$ and $M_n := \frac{1}{n} S_n$. Since $X_j \geq 0$ $\mathbb{P}$ -a.s. for all $j \in \mathbb{N}$, we have $S_n \geq 0$ $\mathbb{P}$ -a.s., so $\sqrt{S_n}$ is well-defined. Moreover, $\sqrt{S_n} + \sqrt{n} \geq \sqrt{n} > 0$, so

we can multiply the numerator and denominator of our target expression by $\sqrt{S_n} + \sqrt{n}$:

$$\begin{aligned}
\frac{2}{\sigma} (\sqrt{S_n} - \sqrt{n}) &= \frac{2}{\sigma} \cdot \frac{(\sqrt{S_n} - \sqrt{n})(\sqrt{S_n} + \sqrt{n})}{\sqrt{S_n} + \sqrt{n}} \\
&= \frac{2}{\sigma} \cdot \frac{S_n - n}{\sqrt{S_n} + \sqrt{n}} \\
&= \frac{2}{\sigma} \cdot \frac{S_n - n}{\sqrt{n} \left(\sqrt{\frac{S_n}{n}} + 1 \right)} \\
&= \frac{S_n - n}{\sigma \sqrt{n}} \cdot \frac{2}{1 + \sqrt{S_n/n}} \\
&= \frac{S_n - n}{\sigma \sqrt{n}} \cdot \frac{2}{1 + \sqrt{M_n}}
\end{aligned}$$

We define

$$U_n := \frac{S_n - n}{\sigma \sqrt{n}}, \quad V_n := \frac{2}{1 + \sqrt{M_n}},$$

so that:

$$\frac{2}{\sigma} (\sqrt{S_n} - \sqrt{n}) = U_n \cdot V_n. \quad (1)$$

Step 2: Central Limit Theorem applied to U_n . Since $\sigma^2 = \text{Var}(X_1) < \infty$, the second moment of X_1 satisfies $\mathbb{E}(X_1^2) = \text{Var}(X_1) + (\mathbb{E}(X_1))^2 = \sigma^2 + 1 < \infty$, so $(X_k)_{k \in \mathbb{N}} \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$. Applying the Central Limit Theorem (Theorem 4.4.1) with $\mu = \mathbb{E}(X_1) = 1$ and $\sigma^2 = \text{Var}(X_1)$, we obtain:

$$U_n = \frac{\sum_{k=1}^n X_k - n \cdot 1}{\sigma \sqrt{n}} \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0, 1). \quad (2)$$

Step 3: Convergence in probability of V_n . We show $V_n \xrightarrow{p} 1$ in three sub-steps.

3.a) $M_n \xrightarrow{p} 1$. Since $(X_k)_{k \in \mathbb{N}} \subset L^1(\Omega, \mathcal{F}, \mathbb{P})$ are i.i.d. with $\mathbb{E}(X_1) = 1$, Kolmogorov's Strong Law of Large Numbers (Theorem 4.3.6) gives:

$$M_n = \frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{\text{a.s.}} \mathbb{E}(X_1) = 1.$$

Since $\mathbb{P}$ -a.s. convergence implies convergence in probability (Theorem 4.1.2 a)), it follows that $M_n \xrightarrow{p} 1$.

3.b) $\sqrt{M_n} \xrightarrow{p} 1$. Since $X_j \geq 0$ $\mathbb{P}$ -a.s., we have $M_n \geq 0$ $\mathbb{P}$ -a.s., so $\sqrt{M_n}$ is well-defined. Moreover, $\sqrt{M_n} \geq 0$ $\mathbb{P}$ -a.s. implies $\sqrt{M_n} + 1 \geq 1$ $\mathbb{P}$ -a.s., and therefore:

$$|\sqrt{M_n} - 1| = \left| (\sqrt{M_n} - 1) \cdot \frac{\sqrt{M_n} + 1}{\sqrt{M_n} + 1} \right| = \frac{|M_n - 1|}{|\sqrt{M_n} + 1|} \leq |M_n - 1|.$$

Then, for any $\varepsilon > 0$, we have $\{|\sqrt{M_n} - 1| > \varepsilon\} \subseteq \{|M_n - 1| > \varepsilon\}$ and, by monotonicity (Lemma 1.1.10d), it then follows that:

$$0 \leq \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right) \leq \mathbb{P} (|M_n - 1| > \varepsilon).$$

The inequality holds for all $n \in \mathbb{N}$, so we can take the limit as $n \rightarrow \infty$:

$$0 \leq \limsup_{n \rightarrow \infty} \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right) \leq \lim_{n \rightarrow \infty} \mathbb{P} (|M_n - 1| > \varepsilon). \quad (3)$$

We know that $M_n \xrightarrow{p} 1$ (from 3a). By the definition of convergence in probability (Definition 4.1.1), $\lim_{n \rightarrow \infty} \mathbb{P}(|M_n - 1| > \varepsilon) = 0$, which, combined with (3), gives:

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right) = 0.$$

Since $0 \leq \liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$ for any non-negative sequence a_n , it follows by the squeeze theorem that both the limit superior and limit inferior are equal to 0. Thus, the full limit exists and:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right) = 0.$$

Therefore, by Definition 4.1.1, $\sqrt{M_n} \xrightarrow{p} 1$.

3.c) $V_n \xrightarrow{p} 1$. Using $1 + \sqrt{M_n} \geq 1$ $\mathbb{P}$ -a.s. (from 3b), we estimate:

$$|V_n - 1| = \left| \frac{2}{1 + \sqrt{M_n}} - 1 \right| = \left| \frac{1 - \sqrt{M_n}}{1 + \sqrt{M_n}} \right| = \frac{|\sqrt{M_n} - 1|}{1 + \sqrt{M_n}} \leq |\sqrt{M_n} - 1|.$$

Then, for any $\varepsilon > 0$, we have $\{|V_n - 1| > \varepsilon\} \subseteq \{|\sqrt{M_n} - 1| > \varepsilon\}$ and, by monotonicity (Lemma 1.1.10d), it then follows that:

$$0 \leq \mathbb{P}(|V_n - 1| > \varepsilon) \leq \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right).$$

The inequality holds for all $n \in \mathbb{N}$, so we can take the limit superior as $n \rightarrow \infty$:

$$0 \leq \limsup_{n \rightarrow \infty} \mathbb{P}(|V_n - 1| > \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \sqrt{M_n} - 1 \right| > \varepsilon \right). \quad (4)$$

We know that $\sqrt{M_n} \xrightarrow{p} 1$ (from 3b). By the definition of convergence in probability (Definition 4.1.1), $\lim_{n \rightarrow \infty} \mathbb{P}(|\sqrt{M_n} - 1| > \varepsilon) = 0$, which, combined with (4), gives:

$$\limsup_{n \rightarrow \infty} \mathbb{P}(|V_n - 1| > \varepsilon) = 0.$$

Since $0 \leq \liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$ for any non-negative sequence a_n , it follows by the squeeze theorem that both the limit superior and limit inferior are equal to 0. Thus, the full limit exists and:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|V_n - 1| > \varepsilon) = 0.$$

Therefore, by Definition 4.1.1, $V_n \xrightarrow{p} 1$.

Step 4: Conclusion via Slutsky's Theorem. We have established $U_n \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$ from (2), and $V_n \xrightarrow{p} 1$ from Step 3. Define the sequence of constant random variables $\tilde{Z}_n \equiv 0$ so that $\tilde{Z}_n \xrightarrow{p} 0$. Applying Slutsky's Theorem with $X_n := U_n \xrightarrow{d} Z$, $Y_n := V_n \xrightarrow{p} 1 =: y$, and $Z_n := \tilde{Z}_n \xrightarrow{p} 0 =: z$, we obtain:

$$\frac{2}{\sigma} \left(\sqrt{S_n} - \sqrt{n} \right) \stackrel{\text{by (1)}}{=} \tilde{Z}_n + U_n \cdot V_n \xrightarrow{d} 0 + 1 \cdot Z = Z.$$

Since $S_n = \sum_{k=1}^n X_k$, we have:

$$\frac{2}{\sigma} \left(\sqrt{\sum_{k=1}^n X_k} - \sqrt{n} \right) \xrightarrow{d} Z \sim \mathcal{N}(0, 1).$$

□